Module (03) :- Frequency Demodulation :-

The process of recovering the original msg s/g from the frequency modulated signal

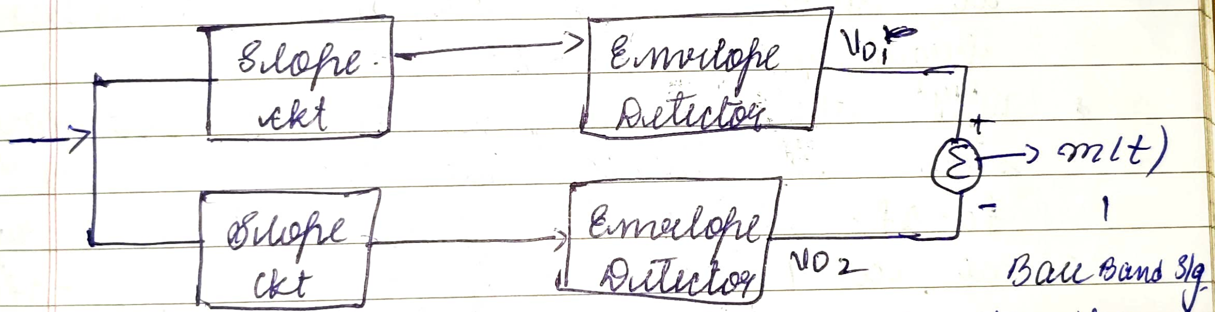
There are 2-types of FM demodulator

i) Frequency discriminator / Balanced slope detector

ii) Phase locked loop. [PLL]

i) Frequency Discriminator [Balanced Slope Detector]

It ~~contains~~ consists of 2 slope detector ckts and block diagram of BSD is as shown.



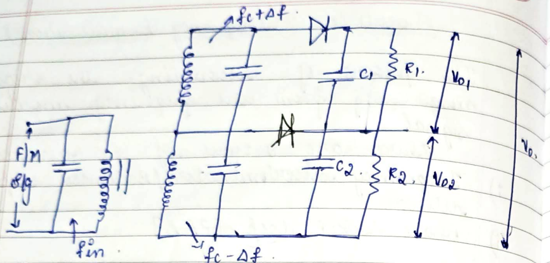
$V_0 = V_{01} - V_{02}$

The output of frequency discriminator is given by.

incoming FM S/g.

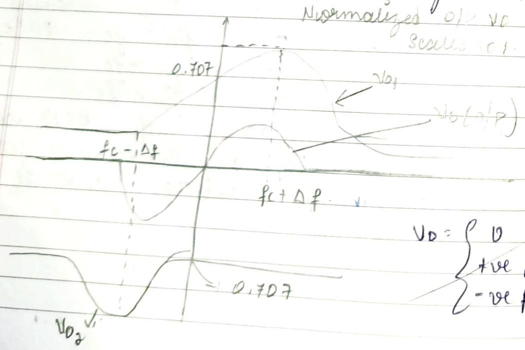
$$V_0 = \begin{cases} 0 & f_{in} = f_c \\ +ve & f_{in} > f_c \\ -ve & f_{in} < f_c \end{cases}$$

The Eq ckt diagram of BSD consists of center tapped transformer followed by 2 Envelope detector and it is shown.



- The FM signal is applied to primary side center tapped transformer and it is tuned to the frequency of a FM S/g
- The upper part of secondary is tuned to frequency $f_c + \Delta f$. hence it gives max response at resonance frequency of $f_c + \Delta f$.
- IIIly the lower part of the secondary side is tuned to the frequency $f_c - \Delta f$. hence it gives minimum response at resonance frequency of $f_c - \Delta f$.

Hence frequency response of balanced slope detector ckt is given below



$$V_o = \begin{cases} 0 & f_{in} = f_c \\ +ve & f_{in} > f_c \\ -ve & f_{in} < f_c \end{cases}$$

→ AN FM wave is defined by $s(t) = 10\cos(2 + \sin 6\pi t)$. Find inst frequency of $s(t)$.

$$s(t) = A_c \cos(\theta_i t)$$

$$f_i(t) = \frac{1}{2\pi}\left[\frac{d}{dt}(\theta_i(t))\right]$$

$$f_i(t) = \frac{1}{2\pi}\left[\frac{d}{dt}(2 + 3\sin 6\pi t)\right]$$

$$= \frac{1}{2\pi}\left[0 + \cos 6\pi t \cdot \overset{3}{\cancel{6\pi}}\right]$$

$$= 3\cos 6\pi t$$

Imp 4 **Super Hetrodyne Receiver :-**

→ In communication system irrespective of ~~what~~ wether it is based on AM or FM the reciever not only had a task demodulating the incoming modulated S/g but it also required to perform the following 3 system functions

i) Carrier frequency tuning
ii) Filtering opr^n
iii) Amplification

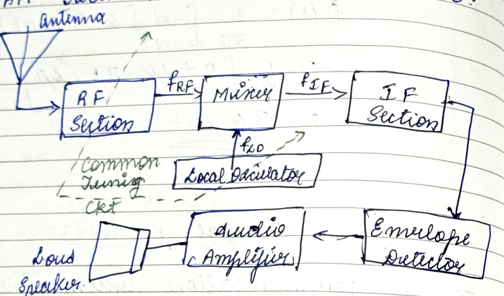
→ A special kind of a receiver that performs the above said three system functions is called as super hetrodyne receiver.

→ It consists of the following section.

i) RF Section
ii) Mixer
iii) Local Oscillator

iv) Intermediate Frequency section
v) Demodulator
vi) Power amplifier

→ Block diagram of super hetrodyne AM receiver is shown below.



→ The incoming AM S/g is picked by up by receiving antenna and provided RF section

→ RF section amplifies the recived S/g i.e., It tuned to the carrier frequency of the incoming ~~signal~~ S/g

→ The O/P of RF section is provided to the mixer ckt along with local oscillator S/g.

→ The combination of mixer and local oscillator provides hetero-dyning func. i.e., the incoming S/g is converted to a intermediate frequency S/g.

→ The intermediate frequency is defined by $f_{IF} = f_{RF} - f_{LO}$

where f_{LO} = Frequency of Local oscillator

f_{RF} = frequency of Incoming RF S/g

→ This resulting IF signal is provided to a IF section which consists of one (or) more stages of tuned amplification. This section provides most of the amplification and selectivity in the receiver.

→ O/P of IF section is applied to the demodulator. The purpose of which is to recover the message S/g.

→ The final operation in the Receiver is the power amplification of Recovered message S/g.

→ Typical frequency parameter of commercial AM and FM radio receiver are listed as follows.

	Parameter	AM radio	FM radio
1	RF carrier Range	0.535 - 1.605 MHz	88 - 108 MHz
2	Mid Band Frequency of IF S/g	455 KHz	10.7 MHz
3	IF Bandwidth	10 KHz	200 KHz

+ Phase Locked Loop [PLL]

→ PLL is a -ve ~~f/b~~ feedback system.

→ It operation is very close to the FM.

→ PLL can be used for.

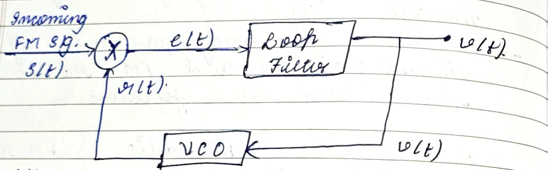
i) Syncronisation

ii) Frequency multiplication

iii) Frequency division
iv) Frequency modulation
v) Indirect method of frequency demodulation

Basically PLL consists of 3 components
i) multiplier
ii) Voltage controlled oscillator
iii) Loop filter

Basic block diagram of PLL is shown below



Initially we set the voltage controlled oscillator so that when the control voltage $v(t) = 0$, two conditions are satisfied

i) Frequency of VCO is same as unmodulated carrier frequency f_c.
ii) The VCO O/P has 90° phase shift w.r.t incoming FM sig.

- The incoming FM sig $s(t)$ is applied to the multiplier along with the VCO O/P $r(t)$.
- The O/P of multiplier is given by
$e(t) = s(t) * r(t)$ ⇒ error signal
where
$s(t)$ and $r(t)$ are defined by
$s(t) = A_c \sin[2\pi f_c t + \phi_1(t)]$
$\phi_1(t) = 2\pi K_f \int_0^t m(t)\, dt$

$r(t) = A_v \cos[2\pi f_c t + \phi_2(t)]$
$\phi_2(t) = 2\pi K_v \int_0^t v(t)\, dt$

where
A_c = peak amplitude of carrier s/g.
A_v = peak amplitude of VCO.
K_f = frequency sensitivity of FM
K_v = frequency sensitivity of VCO.

∴ O/P of multiplier is given by

$e(t) = A_c \sin[2\pi f_c t + \phi_1(t)] \, A_v \cos[2\pi f_c t + \phi_2(t)]$
$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

After simplification we can notice that the multiplier O/P consists of following two terms.

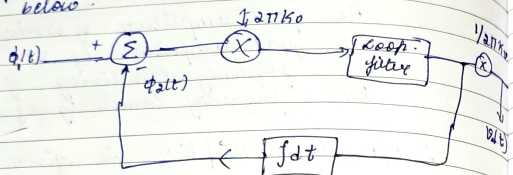
i) Double frequency term.
$K_m A_c A_v \sin[4\pi f_c t + \phi_1(t) + \phi_2(t)]$
ii) Difference frequency term.
$K_m A_c A_v \sin[\phi_1(t) - \phi_2(t)]$
where K_m = multiplier gain

The response of loop filter to HFC is negligibly small. Therefore I/P to the loop filter is reduced to
$e(t) = K_m A_c A_v \sin[\phi_1(t) - \phi_2(t)]$
where. $\phi_1(t) - \phi_2(t) = \phi(t)$ is called as phase error.

- Linear Model of PLL :-

The PLL is said to be phase lo~~t~~ck at when the phase error is zero i.e., $\phi_e(t) = \phi_1(t) - \phi_2(t) = 0$

The linear model of PLL ~~not~~ used for FM demodulation is shown below.



WKT

$$\phi_1(t) = 2\pi k_f \int_0^t m(t)\,dt$$

$$\phi_2(t) = 2\pi k_v \int_0^t v(t)\,dt \quad \text{— ①}$$

$$\phi_e(t) = \phi_1(t) - \phi_2(t) \quad \text{— ②}$$

WKT for a phase locked mode/condn

$$\phi_e(t) = 0$$

$$\therefore \phi_1(t) = \phi_2(t).$$

using the Eqn ① & ② we get.

$$\cancel{2\pi} k_f \int_0^t m(t)\,dt = \cancel{2\pi} k_v \int_0^t v(t)\,dt.$$

Differentiating above eqn w.r.t we get.

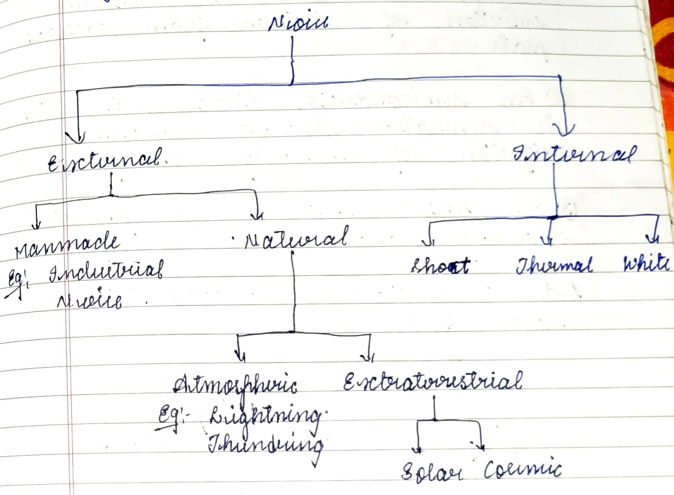
$$k_f\, m(t) = k_v\, v(t).$$

$$\boxed{v(t) = \frac{k_f}{k_v}\, m(t)}$$

Hence O/P $v(t)$ of PLL is proportional to $m(t)$.

Noise and its types

Defn:- It is a disturbance / unwanted frequency s/g appears in or mixed up with wanted frequency s/g



Internal Noise:- It is the one which is generated within the communication system.

→ There are three different types of internal noise.

→ Shot noise

→ Thermal noise

→ White noise

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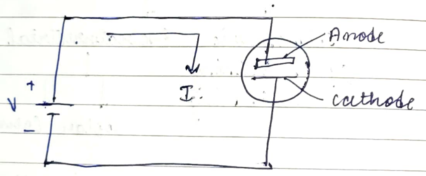
S.M a) Shot Noise :-
Shot noise appears in active devices due to random behaviour of charge carriers such as electrons and holes.

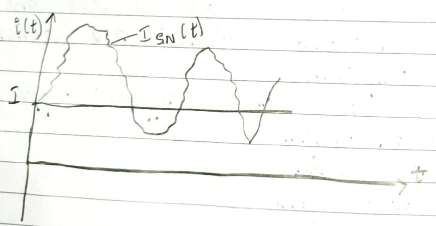
Eg:- ① In vaccum tubes shot noise is generated due to random emission of electrons from cathode plate.

② In semiconductor diode shot noise is generated due to random diffusion of minority charge carriers.

③ In LED's shot noise is generated due to random emission of photons.

Let us consider a vaccum tube as shown below.





The total current in the vaccum

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is given by.

$$i(t) = I + I_{SN}(t)$$

where

I = actual dc current.

$I_{SN}(t)$ = shot noise current.

The mean square value of a ~~flycuating~~ fluctuating shot noise current in vacuum tube is.

$$E[I_{SN}^2] = 2qIB_N.$$

$q = 1.6\times10^{-19}C$.

I = current

B_N = Noise Eq. Bandwidth.

|||ly

For PN Junc diode, mean square value is given by.

$$E[I_{SN}^2] = 2q(I+I_S)B_N.$$

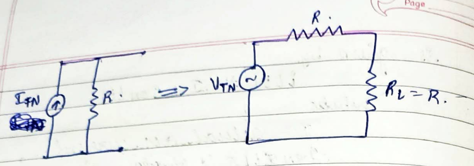
I_S = saturation current which is due to minority charge carriers which doubles for every 10° rise in temp.

b) Thermal Noise :- It appears in ~~ohmic~~ conductors due to random motion of thermally induced electrons.

→ The random motion of thermally induced electrons produced electric current which is random in nature.

→ This random current is called as thermal noise.

→ The below figure shows noise model using resistor.



I_{TN} = Thermal noise current
V_{TN} = Thermal noise voltage $= I_{TN} \times R$

→ This ckt is used to find the maximum noise power
cond^n: $\boxed{R_L = R}$

→ Mean square value of thermal noise voltage $E[V_{TN}^2] = 4KB_NTR$.

where
K = Boltzman const $= 1.38 \times 10^{-23}$ J/sec.
B_N = Noise Eq Bandwidth (Hz)
T = Temperature in Kelvin
R = Resistance in Ω.

→ Maximum noise power produced across R_L is given by

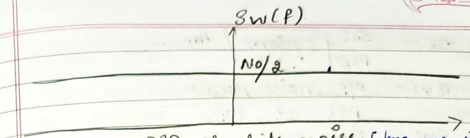
$$P_N = \frac{E[V_{TN}^2]}{4R_L}\Big|_R$$

$$\boxed{P_N = \frac{4KB_NT\cancel{R}}{4\cancel{R}} = KB_NT}$$

5M

c) White Noise:-
White noise has Gaussian distribution and have flat power spectral density ~~range~~ over a frequency range extending from $-\infty$ to ∞ as illustrated below.

→ Represented by $w(t)$

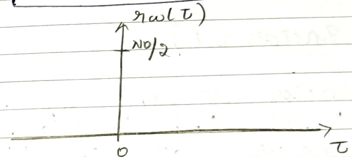


PSD of white noise [frequency domain repr^n]

→ Power spectrum density of white noise is represented by $\boxed{S_w(f) = N_0/2}$ watts/Hz.

→ Autocorrelation is given by. [time domain representation]

$$\boxed{r_w(\tau) = \frac{N_0}{2}\delta(\tau)}$$



→ $S_w(f) \xleftrightarrow{IFT} r_w(\tau)$;

$\frac{N_0}{2} \xleftrightarrow{IFT} \frac{N_0}{2}\delta(\tau)$

2M Noise Eq Bandwidth [B_N]:-

→ It represents the frequency selectivity of filter.
Filter is a frequency selective device

→ Bandwidth is range of frequency over which the system operates.

→ ~~Band~~ Noise Eq Bandwidth is represented by B_N. Mathematically it is defined as follows:
It is the ratio of O/P noise power to a power spectrum density at i/p.

~~B_N~~ S_{ni}, PSD → System → P_{NO}

$$BW = \frac{\text{o/p noise Power}}{\text{PSD at i/p}}$$

$$BW = \frac{P_{NO}}{S_{ni}}$$

$$BW = \frac{KT}{4RC\,KT}$$

The simplified noise Eq Bandwidth Eq^n

$$\boxed{B_N = \frac{1}{4RC}}$$

Noise Factor [F]

Noise factor of any system is defined as the ratio signal to noise power at the i/p to the ratio of signal to noise power at the o/p.

It is defined by

$$F = \frac{(S/N)\ \text{Power at i/p}}{(S/N)\ \text{Power at o/p}}$$

$$\boxed{F = \frac{P_{si}/P_{noi}}{P_{so}/P_{no}}}$$

S+n (P_{si}, P_{ni}) → System → P_{so}, P_{no}

where

P_{si} = Power signal at i/p

P_{noi} = noise ~~signal~~ Power at i/p

P_{so} = Signal Power at o/p

P_{no} = noise Power at o/p

Noise Figure: [FdB]

The representation of Noise Factor in terms of dB is called as noise figure.

$$\boxed{F_{dB} = 10 \log [F]}$$

$$= 10 \log \left[\frac{P_{si}/P_{ni}}{P_{so}/P_{no}}\right]$$

$$= 10 \log \left[\frac{P_{si}}{P_{ni}}\right] - 10 \log \left[\frac{P_{so}}{P_{no}}\right]$$

= (S/N) Power at i/p in dB − (S/N) power at o/p in dB.

Equivalent Noise Temperature [Te]

$$T_e = (F-1)\,T$$

where T_e = Eq. noise temperature

F = noise factor

T = Temp in Kelvin.

Noise Eq. B.W :-

$$B_N = \frac{1}{4RC}$$

Mean Square value of shot noise current

$$E[I^2_{SN}] = 2 I q B_N$$

Mean Square value of thermal noise voltage

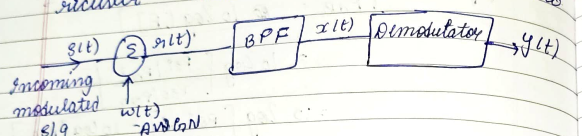
$$E[V^2_{TN}] = 4KTB_N R.$$

Noise Power :-

$$P_N = KTB_N$$

Noise in CW modulation System

To study the noise in CW modulation system, let us consider the basic receiver model as illustrated below



where s(t) represents Incoming modulated s/g
w(t) represents white Gaussian noise occurs at front end of receiver
r(t) is the received noisy s/g

$r(t) = s(t) + w(t)$

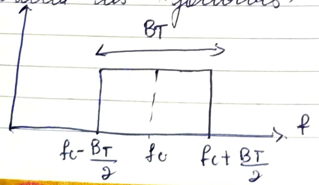
→ This noisy s/g is applied to the Band Pass Filter which performs two basic operations

i) Filtering

ii) Amplification

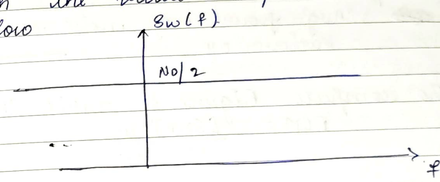
→ The bandwidth of BandPass Filter is Just Enough to pass modulated s/g s(t). Hence the Bandwidth of a BandPass Filter is Equal to Bandwidth of a modulated s/g s(t) and the midband frequency is Equal to carrier frequency f_c.

∴ The frequency response of this filter is illustrated as follows.



The ~~demoduna~~ demodulator used in this model depends on the type of a modulation used.

→ To perform a noise analysis of a communication s/m we assume that the noise component w(t) is AWGN having flat power spectrum density with the value $N_0/2$ as illustrated below



→ The BandPass filter O/P is defined by

~~$x(t) = s(t) + w(t)$~~

$x(t) = s(t) + n(t)$.

where

$n(t)$ = Filtered Narrow Band Noise component.

∴ Filtered noise component n(t) is Narrow Band noise. Hence it is represented in Canonical form as follows.

$n(t) = n_I(t) \cos 2\pi f_c t - n_Q(t) \sin 2\pi f_c t$

$n_I(t)$ = Inphase component

$n_Q(t)$ = Quadrature component

After performing the filtering opr^n we need to find avg power of a modulated s/g s(t) at demodulator i/p and avg power of filtered n(t) at demodulator i/p.

→ Signal to noise power at the

demodulator i/p is computed as follows

$$[SNR]_{I/P} = \frac{\text{Avg Power of } s(t)}{\text{Avg Power of } n(t)} \quad \text{—— ①}$$

IIIly, inorder to compute the o/p SNR of reciever,

$$[SNR]_{o/p} = \frac{\text{Avg power of demodulated message s/g } m(t) \text{ at reciever o/p}}{\text{Avg power of noise comp at reciever o/p}} \quad \text{—— ②}$$

To compute Figure of Merit [FOM].

$$FOM = \frac{[SNR]_{I/P}}{[SNR]_{o/p}}.$$

→ Higher the value of FOM better the performance of the Reciever and lower the value of FOM lower the performance of reciever.

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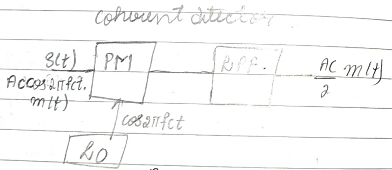
10m

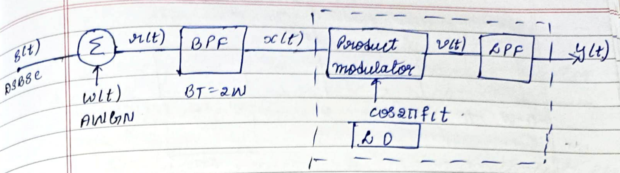
Noise in DSBSC Reciever model:-

ST FOM of DSBSC is 1

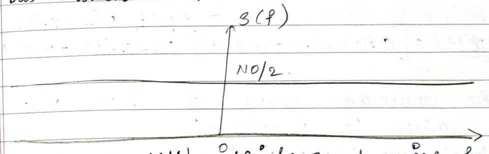
coherent detector



The reciever block diagram of DSBSC using coherent detector is shown below.



In this reciever the incoming $s(t)$ is DSBSC s/g and it is defined by $s(t) = A_c \cos 2\pi f_c t \, m(t)$ and $w(t)$ represents Additive White Gaussian Noise having power spectral density as illustrated below.



$w(t)$ = Wide Band noise

→ The recieved noisy s/g $r(t)$ is defined by $r(t) = s(t) + w(t)$

$$r(t) = A_c \cos 2\pi f_c t . m(t) + w(t)$$

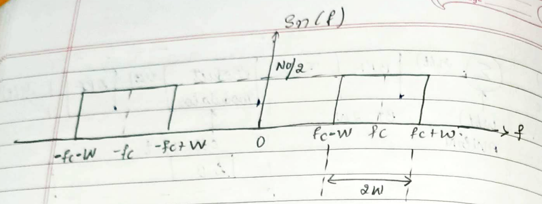
→ The recieved noisy s/g $r(t)$ is applied to the BPF with the BW $B_T = 2W$ Hz i.e., Band width of a DSBSC modulated s/g. Hence BPF passes DSBSC s/g $s(t)$ without any disturbance. The filtered o/p $x(t)$ is represented as

$$x(t) = C A_c \cos 2\pi f_c t . m(t) + n(t)$$

where

C = System dependent scaling factor

$n(t)$ = narrow band noise component having PSD as illustrated below.



Since n(t) is a narrow band component it can be represented in canonical form as follows.

$$n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$$

$\therefore x(t) \Rightarrow$

$$x(t) = C A_c \cos 2\pi f_c t \cdot m(t) + n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$$

$$x(t) = [C A_c m(t) + n_I(t)]\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$$

→ The average power of modulated s/g [s(t)] is given as follows.

Avg power of s(t) $= \dfrac{C^2 A_c^2}{4} \cdot P$

$P_c = \dfrac{V_{rms}^2}{2}$

$V_{rms} = \dfrac{V_m}{\sqrt{2}} = \dfrac{C A_c}{\sqrt{2}}$

$P_c = \dfrac{C^2 A_c^2}{4}$

Avg power of NB n(t) $= \dfrac{N_o}{\cancel{2}} \cancel{2}W = N_o W$

I/P SNR is given by

$$[SNR]_{I/P} = \frac{\text{Avg Power of } s(t) \text{ at demodulator I/P}}{\text{Avg power of } n(t) \text{ at detector I/P}}$$

$$= \frac{\frac{C^2 A_c^2 P}{4}}{N_o W}$$

$$= \frac{C^2 A_c^2 P}{4 N_o W} \quad \text{—— (1)}$$

The s/g $s(t) = [C A_c m(t) + n_I(t)]\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$ is applied to the coherent detector and it produces a o/p

$$y(t) = \frac{C A_c m(t) + n_I(t)}{2}$$

The avg power of demodulated s/g m(t) at receiver O/P is

$$O/P = \frac{C^2 A_c^2 P}{4}$$

Avg Power of noise component at receiver o/p is $= N_o/2 \cdot W$

O/P SNR is defined by.

$$[SNR]_{o/p} = \frac{\text{Avg Power of demodulated s/g } m(t) \text{ at receiver o/p}}{\text{Avg power of noise component at receiver o/p}}$$

$$= \frac{\frac{C^2 A_c^2 P}{4}}{\frac{N_o W}{2}}$$

$$[SNR]_{o/p} = \frac{C^2 A_c^2 P}{2 N_o W} \quad \text{—— (2).}$$

W.K.T

$$[FOM] = \frac{[SNR]_O}{[SNR]_I} = \frac{\frac{C^2 A_c^2 P}{2 N_o W}}{\frac{C^2 A_c^2 P}{2 N_o W}}$$

$$\boxed{[FOM] = 1}$$